



Wk1: Data Wrangling Problem



Rojesh Shikhrakar • 27 Apr (Edited 09:34)

Assignments

Due 4 May, 23:59

• 100 points

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Each week, we will release a new problem set to be discussed during our upcoming session. To get the most out of our time together, **you are expected to complete these tasks before the session begins.**

Guidelines for Success:

- **Document Everything:** Journal your failures, breakthroughs, and key learnings as you work.
- **Leverage AI:** You are encouraged to utilize AI tools to assist with your coding process.

This Week's Tasks:

1) Data Wrangling : Data Scientist at a Regional Health Center

Objective: Clean and prep patient data from three separate systems (Administrative, Lab, and Lifestyle) to ensure it is "analysis-ready."

Key Tasks:

- **Structure & Quality:** Inspect the three provided datasets.
- **Cleaning:** Handle missing values and anomalies.
- **Standardization:** Fix data formats (e.g., Blood Pressure strings).
- **Documentation:** Justify your decisions within the notebook.

Requirements:

- **Think Critically:** There is no single "right" answer—justify your cleaning choices.
- **AI-Assisted:** You are encouraged to use AI to optimize your code.
- **Journal:** Document your failures and breakthroughs.



Deadline: Please complete this notebook **before** our next session.

Files: See the attached .ipynb and .csv files below. 🚀

2) Assignment: SQL Fundamentals: Practice cleaning, joining, and aggregating data using the **Classic Models** retail database.

🔧 **Getting Started: Import the DB:** Use the source command in MySQL to load the provided .sql file.

- *Example:* `mysql> source C:/path/to/mysqlsampledatabase.sql;`

Explore the Schema: Familiarize yourself with the 8 core tables (Customers, Products, Orders, etc.).

✅ **Key Tasks:**

- **Transform:** Use CASE statements and filtering to clean data.
- **Join:** Connect customers to their orders and payments.
- **Aggregate:** Calculate business metrics like total sales and employee performance.

📄 **Expectations:**

- **Complete before the next session.**
- Log your **SQL errors and solutions** in your learning journal.
- Feel free to use **AI** to debug complex queries.

Files: Download the `mysqlsampledatabase.sql` file attached in the doc.

<u>Wk_1_Data_W...</u> Unknown		<u>clinical_data.c...</u> CSV	
<u>lifestyle_facto...</u> CSV		<u>patient_demo...</u> CSV	
<u>SQL Assignme...</u> Microsoft Word		<u>mysqlsampled...</u> SQL	

Your work

Assigned 

+ Add or create

Mark as Done

Work cannot be turned in after the due date

 Private comments

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